01 Preprocessing

2026-03-13

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Setup

Read GDC cart metadata JSON, clean into a dataframe, and save to `Data/Input/`.

```
if (!exists("project_root")) {
    cur <- tryCatch(knitr::current_input(), error = function(e) NULL)
    path <- if (is.null(cur))
        NULL else if (is.character(cur))
        cur else cur[["path"]]
    if (!is.null(path) && length(path) > 0) {
        project_root <- normalizePath(file.path(dirname(path[1]),
            ".."), mustWork = FALSE)
    } else {
        wd <- getwd()
        project_root <- if (dir.exists(file.path(wd, "Data")))
            wd else normalizePath(file.path(wd, ".."), mustWork = FALSE)
    }
}
DATA_DIR <- file.path(project_root, "Data")
INPUT_DIR <- file.path(DATA_DIR, "Input")
RESULTS_DIR <- file.path(project_root, "Results", "01_Preprocessing")
METADATA_JSON <- file.path(DATA_DIR, "metadata.cart.2026-03-12.json")
CLINICAL_TSV <- file.path(DATA_DIR, "clinical.cart.2026-03-12",
    "clinical.tsv")
```

```r
SAMPLE_TSV <- file.path(DATA_DIR, "biospecimen.cart.2026-03-12",
    "sample.tsv")
METADATA_RDS <- file.path(RESULTS_DIR, "metadata.rds")
CLINICAL_RDS <- file.path(RESULTS_DIR, "clinical.rds")
SAMPLE_RDS <- file.path(RESULTS_DIR, "sample.rds")
COUNT_MATRIX_RDS <- file.path(INPUT_DIR, "count_matrix.rds")
```

Packages

```r
library(jsonlite)
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:dbplyr':
##
##     ident, sql

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tibble)
```

Helper for missing JSON fields

```r
`%||%` <- function(x, y) if (is.null(x)) y else x
```

Read metadata JSON

```r
raw <- read_json(METADATA_JSON, simplifyVector = FALSE)
length(raw)
```

```
## [1] 1224
```

Clean into a dataframe

Flatten top-level file fields and the first `associated_entity` (case/aliquot) per file. Nested `analysis` and `annotations` are omitted for a simple table.

```r
rows <- lapply(raw, function(x) {
    ae <- if (length(x$associated_entities) > 0)
        x$associated_entities[[1]] else list()
    data.frame(file_id = x$file_id %||% NA_character_, file_name = x$file_name %||%
        NA_character_, submitter_id = x$submitter_id %||% NA_character_,
        data_format = x$data_format %||% NA_character_, data_category = x$data_category %||%
```

```r
            NA_character_, data_type = x$data_type %||% NA_character_,
        file_size = as.integer(x$file_size %||% NA_integer_),
        md5sum = x$md5sum %||% NA_character_, state = x$state %||%
            NA_character_, platform = x$platform %||% NA_character_,
        experimental_strategy = x$experimental_strategy %||%
            NA_character_, access = x$access %||% NA_character_,
        case_id = ae$case_id %||% NA_character_, entity_submitter_id = ae$entity_submitter_id %||%
            NA_character_, entity_type = ae$entity_type %||%
            NA_character_, entity_id = ae$entity_id %||% NA_character_,
        workflow_type = (if (!is.null(x$analysis))
            x$analysis$workflow_type else NULL) %||% NA_character_, stringsAsFactors = FALSE)
})

metadata <- bind_rows(rows) %>%
    as_tibble() %>%
    mutate(sample_submitter_id = substr(entity_submitter_id,
        1, 16))

dim(metadata)
```

```
## [1] 1224   18
```

```r
head(metadata)
```

```
## # A tibble: 6 x 18
##   file_id   file_name submitter_id data_format data_category data_type
##   <chr>     <chr>     <chr>        <chr>       <chr>         <chr>
## 1 744a6d3d~ 94027f46~ 41848a66-83~ TSV         Transcriptom~ Gene Exp~
## 2 4ecc1f1a~ df45fb41~ 25982983-a9~ TSV         Transcriptom~ Gene Exp~
## 3 1ace2a0c~ c33ecc28~ 0493c6f8-6f~ TSV         Transcriptom~ Gene Exp~
## 4 2d5b0962~ 602e3d24~ 40157473-bf~ TSV         Transcriptom~ Gene Exp~
## 5 84cff855~ 03d891b3~ 18a83f64-97~ TSV         Transcriptom~ Gene Exp~
## 6 c2515ef2~ f8ff90c6~ 43c43fde-1f~ TSV         Transcriptom~ Gene Exp~
## # i 12 more variables: file_size <int>, md5sum <chr>, state <chr>,
## #   platform <chr>, experimental_strategy <chr>, access <chr>,
## #   case_id <chr>, entity_submitter_id <chr>, entity_type <chr>,
## #   entity_id <chr>, workflow_type <chr>, sample_submitter_id <chr>
```

Read clinical TSV, keep selected columns, rename, and remove duplicates

```r
clinical_raw <- read.delim(CLINICAL_TSV)
clinical <- clinical_raw %>%
  select(
    project.project_id,
    cases.case_id,
    cases.submitter_id,
    demographic.age_at_index,
    demographic.gender,
    demographic.race,
    demographic.vital_status,
    diagnoses.ajcc_pathologic_stage,
    diagnoses.ajcc_pathologic_t
```

```r
  ) %>%
  rename(
    project_id           = project.project_id,
    case_id              = cases.case_id,
    submitter_id         = cases.submitter_id,
    age         = demographic.age_at_index,
    gender               = demographic.gender,
    race                 = demographic.race,
    vital_status         = demographic.vital_status,
    ajcc_pathologic_stage = diagnoses.ajcc_pathologic_stage,
    ajcc_pathologic_t    = diagnoses.ajcc_pathologic_t
  ) %>%
  mutate(has_missing = if_any(everything(), ~ . == "'--")) %>%
  group_by(case_id) %>%
  filter(
    n() == 1L |                                                    # case_id occurs once: keep it
    (n() > 1L & !has_missing) |                                   # case_id occurs multiple times: ke
    (n() > 1L & sum(!has_missing) == 0L & row_number() == 1L)        # case_id occurs multiple times an
  ) %>%
  ungroup() %>%
  select(-has_missing) %>%
  distinct()
message("Clinical: ", nrow(clinical_raw), " rows → ", nrow(clinical), " after selecting columns, cleani
```

```
## Clinical: 5529 rows → 1115 after selecting columns, cleaning duplicated case_id (drop '-- when possi
```

Read sample TSV

```r
sample <- read.delim(SAMPLE_TSV) %>%
    dplyr::select(project.project_id, cases.case_id, samples.tissue_type,
        samples.submitter_id) %>%
    rename(project_id = project.project_id, case_id = cases.case_id,
        tissue_type = samples.tissue_type, sample_submitter_id = samples.submitter_id) %>%
    filter(sample_submitter_id %in% metadata$sample_submitter_id)
saveRDS(sample, SAMPLE_RDS)

# Add clinical into sample so result has one row per sample
# (same nrow as sample)
clinical_one_per_case <- clinical %>%
    distinct(case_id, project_id, .keep_all = TRUE)
clinical <- sample %>%
    left_join(clinical_one_per_case, by = c("case_id", "project_id"))

# Add metadata into clinical
clinical_one_per_case <- clinical %>%
    distinct(case_id, project_id, .keep_all = TRUE)
metadata_caseonly = metadata %>%
    select(entity_submitter_id, case_id)
clinical <- metadata_caseonly %>%
    left_join(clinical_one_per_case, by = c("case_id"))
```

Add tissue_type from sample into metadata

```
sample <- sample %>%
    select(sample_submitter_id, tissue_type) %>%
    distinct(sample_submitter_id, .keep_all = TRUE)
metadata <- metadata %>%
    left_join(sample, by = "sample_submitter_id")
```

Read count matrix and rename columns to entity_submitter_id

```
count_matrix <- readRDS(COUNT_MATRIX_RDS)
metadata$file_id_clean = gsub(".rna_seq.augmented_star_gene_counts.tsv",
    "", metadata$file_name)
idx <- match(colnames(count_matrix), metadata$file_id_clean)
colnames(count_matrix) <- ifelse(is.na(idx), colnames(count_matrix),
    metadata$entity_submitter_id[idx])
count_matrix[1:5, 1:5]
```

```
##                    TCGA-E9-A1NF-11A-73R-A14D-07
## ENSG00000000003.15                         3076
## ENSG00000000005.6                          2185
## ENSG00000000419.13                         1041
## ENSG00000000457.14                          748
## ENSG00000000460.17                          292
##                    TCGA-OL-A5RV-01A-12R-A28M-07
## ENSG00000000003.15                         3888
## ENSG00000000005.6                             9
## ENSG00000000419.13                         1458
## ENSG00000000457.14                         2135
## ENSG00000000460.17                          440
##                    TCGA-BH-A0HY-01A-11R-A056-07
## ENSG00000000003.15                         1372
## ENSG00000000005.6                            27
## ENSG00000000419.13                         2135
## ENSG00000000457.14                         2385
## ENSG00000000460.17                         1075
##                    TCGA-W8-A86G-01A-21R-A36F-07
## ENSG00000000003.15                         4914
## ENSG00000000005.6                            19
## ENSG00000000419.13                         1507
## ENSG00000000457.14                         1404
## ENSG00000000460.17                          348
##                    TCGA-A0-A12G-01A-11R-A10J-07
## ENSG00000000003.15                         5055
## ENSG00000000005.6                             6
## ENSG00000000419.13                         1849
## ENSG00000000457.14                         1694
## ENSG00000000460.17                          393
```

Save to Results/01_Preprocessing

```r
dir.create(RESULTS_DIR, showWarnings = FALSE, recursive = TRUE)
saveRDS(metadata, METADATA_RDS)
saveRDS(clinical, CLINICAL_RDS)
saveRDS(count_matrix, file.path(RESULTS_DIR, "count_matrix.rds"))
message("Saved to ", RESULTS_DIR)
```

```
## Saved to /Users/xliu2942/Documents/Projects/Tutorial/Results/01_Preprocessing
```

```r
message("  ", METADATA_RDS)
```

```
##   /Users/xliu2942/Documents/Projects/Tutorial/Results/01_Preprocessing/metadata.rds
```

```r
message("  ", CLINICAL_RDS)
```

```
##   /Users/xliu2942/Documents/Projects/Tutorial/Results/01_Preprocessing/clinical.rds
```

```r
message("  ", file.path(RESULTS_DIR, "count_matrix.rds"))
```

```
##   /Users/xliu2942/Documents/Projects/Tutorial/Results/01_Preprocessing/count_matrix.rds
```

Done.

- Metadata: 1224 rows x 20 columns -> `Results/01_Preprocessing/metadata.rds`
- Clinical: 1224 rows -> `Results/01_Preprocessing/clinical.rds`
- Count matrix: 60660 genes x 1224 samples (cols = entity_submitter_id) -> `Results/01_Preprocessing/count_matrix.`